

## EXERCISE 3 : SQL CASE statements

### Updated Exercises

#### 1. Table : products

```
SELECT product_name,  
       price,  
CASE  
  WHEN price > 1000 THEN 'Expensive'  
  WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'  
  WHEN price < 100 THEN 'Budget'  
END AS price_category  
FROM products;
```

↓

| product_name | price   | price_category |
|--------------|---------|----------------|
| Laptop       | 1200.00 | Expensive      |
| Phone        | 800.00  | Mid-range      |
| Keyboard     | 45.00   | Budget         |
| Monitor      | 300.00  | Mid-range      |
| Mouse        | 25.00   | Budget         |

#### 2. Table : orders

```
SELECT customer_name,  
       amount,  
CASE  
  WHEN amount >= 1000 THEN 'High Value'  
  WHEN amount BETWEEN 500 AND 999.99 THEN  
    'Medium Value'  
  WHEN amount < 500 THEN 'Low value'  
END AS order_value_category  
FROM orders;
```

| customer_name | amount  | order_value_category |
|---------------|---------|----------------------|
| Alice         | 150.00  | Low Value            |
| Bob           | 560.00  | Medium Value         |
| Charlie       | 999.99  | Medium Value         |
| Diana         | 45.50   | Low Value            |
| Ethan         | 1200.00 | High Value           |

### 3. Table : employees

```

SELECT emp_name,
       department,
       salary,
CASE
  WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
  WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
  ELSE 'staff'
END AS position_level
FROM employees ;

```



| emp_name | department | salary | position_level |
|----------|------------|--------|----------------|
| John     | IT         | 85000  | Senior IT      |
| Sara     | HR         | 60000  | Experienced HR |
| Mark     | IT         | 75000  | staff          |
| Lucy     | Finance    | 95000  | staff          |
| Tom      | HR         | 55000  | staff          |



#### 4. Table : students

```

SELECT student_name,
       score,
       CASE
         WHEN score >= 90 THEN 'A'
         WHEN score BETWEEN 80 AND 89 THEN 'B'
         WHEN score BETWEEN 70 AND 79 THEN 'C'
         WHEN score BETWEEN 60 AND 69 THEN 'D'
         WHEN score < 60 THEN 'F'
       END AS grade
FROM students;

```



| Student_name | score | grade |
|--------------|-------|-------|
| Anna         | 92    | A     |
| Ben          | 76    | C     |
| Cara         | 59    | F     |
| David        | 83    | B     |
| Ella         | 68    | D     |

#### 5. Table : deliveries

```

SELECT delivery_id,
       delivery_time_minutes,

```

```

CASE

```

```

  WHEN delivery_time_minutes <= 30 THEN 'FAST'
  WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'OnTime'
  WHEN delivery_time_minutes > 60 THEN 'Late'

```

```

END AS performance

```

```

FROM deliveries;

```

| delivery_id | delivery_time_minutes | performance |
|-------------|-----------------------|-------------|
| 1           | 45                    | On Time     |
| 2           | 80                    | Late        |
| 3           | 30                    | Fast        |
| 4           | 65                    | Late        |
| 5           | 100                   | Late        |

## 6. Table : tickets

SELECT issue\_type,  
priority,

CASE

WHEN priority = 3 THEN 'High'

WHEN priority = 2 THEN 'Medium'

WHEN priority = 1 THEN 'Low'

END AS priority\_label

FROM tickets;



| issue_type     | priority | priority_label |
|----------------|----------|----------------|
| Login issue    | 1        | Low            |
| Server down    | 3        | High           |
| Slow system    | 2        | Medium         |
| Email error    | 2        | Medium         |
| Password reset | 1        | Low            |



### 7. Table : attendance

```
SELECT student_id,  
       ((days_present * 100) / total_days) AS attendance_percentage,  
       CASE  
         WHEN ((days_present * 100) / total_days) >= 90 THEN 'Excellent'  
         WHEN ((days_present * 100) / total_days) BETWEEN 75 AND 89  
           THEN 'Good'  
         WHEN ((days_present * 100) / total_days) < 75 THEN  
           'Needs Improvement'  
       END AS attendance_status  
FROM attendance;
```



| Student_id | attendance_percentage | attendance_status |
|------------|-----------------------|-------------------|
| 1          | 90                    | Excellent         |
| 2          | 60                    | Needs Improvement |
| 3          | 96                    | Excellent         |
| 4          | 50                    | Needs Improvement |
| 5          | 100                   | Excellent.        |

### 8. Table : products\_inventory

```
SELECT product_id,  
       stock_qty,  
       CASE  
         WHEN stock_qty = 0 THEN 'Out of Stock'  
         WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'  
         WHEN stock_qty > 5 THEN 'In Stock'  
       END AS stock_status  
FROM products_inventory;
```

| product_id | Stock_qty | stock_status |
|------------|-----------|--------------|
| 1          | 5         | In Stock     |
| 2          | 0         | Out of Stock |
| 3          | 25        | In Stock     |
| 4          | 10        | In Stock     |
| 5          | 3         | Low Stock    |

### 9. Table : classes

```

SELECT subject,
       enrolled_students,
       CASE
         WHEN enrolled_students >= 25 THEN 'Large'
         WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
         ELSE 'Small'
       END AS class_size_category
FROM classes ;

```



| Subject | enrolled_students | class_size_category |
|---------|-------------------|---------------------|
| Math    | 30                | Large               |
| English | 25                | Large               |
| Science | 15                | Medium              |
| Art     | 5                 | Small               |
| History | 20                | Medium              |



10. Table : payments

```
SELECT payment_id,  
        payment_method,  
        amount,
```

```
CASE
```

```
    WHEN payment_method = 'Cash' AND amount >= 200
```

```
        THEN 'Eligible for Discount'
```

```
    ELSE 'Not Eligible'
```

```
END AS discount_eligibility
```

```
FROM payments ;
```



| payment_id | payment_method | amount | discount_eligibility  |
|------------|----------------|--------|-----------------------|
| 1          | Card           | 50.00  | Not Eligible          |
| 2          | Cash           | 200.00 | Eligible for Discount |
| 3          | Card           | 150.00 | Not Eligible          |
| 4          | PayPal         | 75.00  | Not Eligible          |
| 5          | Cash           | 300.00 | Eligible for Discount |